

Idle Vehicle Simulator — Game Design Requirements

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Overview

Idle Vehicle Simulator is a Roblox idle game where everything feeds one number: **Speed**. Players earn it by idling in their workshop, racing NPCs for money, rolling for faster cars, and rebirthing for bigger multipliers.

Design pillars

- **Speed is the score.** Every system either raises Speed gain or is unlocked by it.
- **Progress while idle.** The workshop and auto-rolling keep paying out while the player is AFK.
- **Push your luck.** After each race win, players bank the money or risk it on a harder race.
- **Resets that feel huge.** Each rebirth wipes Money and Speed but makes the next run dramatically faster.

flowchart LR

```

W[Workshop - AFK] -->|Speed/sec| S((Speed))
S --> R[Race NPC ladder]
R -->|cash out| M[Money]
M --> U[Fuel, auras, spoilers,<br/>workshop upgrades]
U -->|multipliers| S
S -->|XP| L[Level]
L -->|milestone| RB[Rebirth]
RB -->|multis + stat points| S
G[Auto-roll cars] -->|car multi| S

```

The loop: idle for Speed, race to turn Speed into Money, spend Money on multipliers, then rebirth once you hit a level milestone.

Glossary

Plain-language definitions for every term used below.

Term	What it means
Speed	The main progression number. Think of it as the player's score.
Speed gain	How much Speed the player earns each second.
Multiplier (multi)	A number gains get multiplied by. 2x doubles, 10x makes it ten times bigger.
Base car	The car the player has equipped. Its multi boosts Speed gain.
Mechanical Workshop	The player's personal area at their base. Standing in it earns Speed passively.
NPC	A computer-controlled racer with a fixed speed.
Luck	Raises the odds of rolling rarer cars. 1x is normal odds.
Auto-roll	The game rolls for a new car on a timer, no input needed.
Jett Fuel	A power that gives every Nth auto-roll a big luck boost.
Rebirth	Resetting Money and Speed in exchange for permanent multipliers and Rebirth Points.
Rebirth Points	Points spent on four permanent stats.
Server Luck	A temporary luck boost for everyone in the server, bought with Robux.
Gamepass	A one-time Robux purchase the player owns forever.
Developer product	A Robux purchase that can be bought again and again (boosts, consumables).

Core stat model

Proposed: every source is a multiplier, and Speed gain per second is all of them multiplied together. The brainstorm doesn't say how they stack, so this is a starting point to confirm.

$$\text{Speed gain per second} = B \times C \times F \times A \times P \times W \times R \times S \times G$$

Symbol	Multiplier	Comes from	Starts at	On rebirth
B	Base rate	Fixed game constant	1 Speed/sec	Unchanged
C	Car	Equipped car from the gacha	1x (starter car)	Kept (proposed)
F	Fuel	Bought with Money	1x	Open question
A	Aura	Bought with Money	1x	Open question
P	Spoiler	Bought with Money	1x	Open question
W	Workshop	Money upgrades + Gold/Diamond/Void passes. Only applies inside the workshop	1x	Passes kept; money upgrades open
R	Rebirth	Automatic bonus each rebirth	1x	Grows
S	Speed stat	Rebirth Points spent on Base Speed Multi	1x	Kept
G	Gamepass	2x Speed pass chain	1x	Kept

In races, Speed gain uses the same formula without W (proposed). Confirm whether races should get every other multiplier or only a flat base rate.

Balance tip: with nine multipliers stacked, a small change to one moves everything. Model the numbers in a spreadsheet before building. Stat points are easier to tune if each point *adds* to its multi (e.g. +0.1x per point) rather than multiplying it.

Starting state and Mechanical Workshop

New players start with the starter car (1x), 0 Speed, 0 Money, level 1, 1x luck and a tier-1 workshop. The workshop is their main source of Speed.

Workshop requirements

- Every player gets their own workshop at their base. Server size should match the number of base plots (e.g. 8 plots, 8 players).

- Speed is earned only while the player stands inside the workshop zone. Leaving the zone stops the gain.
- Gain per second uses the full formula above, including the workshop multi (W).
- Money upgrades raise W in big jumps. Placeholder: each level doubles W and costs 5x the last.
- The Gold, Diamond and Void gamepass tiers add a larger permanent W boost (see Monetization).

Heads-up on AFK: Roblox disconnects players after 20 minutes with no input ([DevForum](#)). A game built on idling needs a plan for that:

- **Offline earnings (recommended):** on rejoin, credit part of what the player would have earned while away. Placeholder: 25% of workshop gain, capped at 8 hours.
- **Welcome-back screen:** show "You were away 3h 12m, collect 4.2M Speed" so the kick feels like a reward, not a loss.

Racing (core game loop)

Racing turns Speed into Money: climb a ladder of NPC races, then bank the pot or risk it on a harder race.

Requirements

- Several environments (placeholder names: City, Desert, Snow, Volcano, Space). Each one is a ladder of tracks, like levels.
- Each track has an NPC with a fixed speed. Every next track's NPC is much faster.
- After a win, the player picks **Cash out** (take the money, go home) or **Continue** (race the next track).
- A loss wipes all unbanked money from that run.
- The payout grows with the difficulty of the track beaten.
- The player keeps earning Speed during races (see Core stat model).

```
stateDiagram-v2
    [*] --> Race
    Race --> Won: beat the NPC
    Race --> Lost: NPC wins
    Won --> CashOut: bank the pot
    Won --> Race: continue, harder NPC
    Lost --> Home: pot lost
    CashOut --> Home
```

Each win raises the pot; each Continue puts the whole pot at risk.

Sample ladder for one environment (placeholder numbers)

Track	NPC speed	Take-home if you cash out here
1	100	\$50
2	500	\$300
3	2.5K	\$2K
4	12.5K	\$12K
5	62.5K	\$75K

NPC speed rises 5x per track and payout about 6x, so pushing further pays off more than it costs.

Design flag: make Continue a real gamble. If a race is just "player Speed beats NPC speed" and both numbers are visible, players always know the result in advance. Continue is never risky. Options:

1. **Win chance from the speed ratio (recommended).** Being faster makes a win likely, not certain. Luck could nudge the odds.
2. **Hide the next NPC's speed.** Show a range or "???" until the race starts.
3. **Light skill check.** A boost button with a timing window.

For option 1, one simple formula (S_p = player speed, S_n = NPC speed, k = tuning knob):

$$P(\text{win}) = \frac{S_p^{\backslash,k}}{S_p^{\backslash,k} + S_n^{\backslash,k}}$$

With $k = 2$, twice the NPC's speed wins 80% of the time; ten times wins 99%.

Level system

Players earn XP every second based on their Speed, so faster players level up faster. Levels matter because they unlock rebirths.

- XP per second scales with current Speed.
- Each level needs more XP than the last.
- Level milestones unlock rebirths (see next section).

Speed grows by multiplying, so XP taken straight from Speed would make late-game levels fly by. Proposed: feed XP from a slower curve, like the square root of Speed. Placeholder formulas:

```
\text{XP per second} = c \times \sqrt{\text{Speed}} \quad \text{XP to next level} = 100 \times 1.15^{\text{level}}
```

Rebirth system

Rebirth trades the player's Money and Speed for a big permanent boost to Speed multi and Luck multi, plus Rebirth Points to spend on stats.

Requirements

- Rebirth unlocks at level milestones: 10, 25, 50, 75, 100, then every 25 after (proposed).
- Rebirthing resets Money and Speed.
- Each rebirth gives a large automatic boost to Speed multi (R) and Luck multi.
- Each rebirth also gives Rebirth Points to spend on four stats.

Proposed milestone rule: if level resets on rebirth, each rebirth needs the next milestone. Rebirth 1 at level 10, rebirth 2 at level 25, and so on. That gives rebirths a rising cost.

Stats (bought with Rebirth Points)

Stat	What it boosts	Per point (placeholder)
Base Speed Multi	All Speed gain	+0.1x
Base Money Multi	Race payouts	+0.1x
Luck Multi	Car roll odds	+0.05x
Mechanical Workshop Multi	Workshop gain only, on top of W	+0.2x

Speed and Luck grow two ways: the automatic rebirth bonus and stat points. That's fine, but the rebirth screen should show both so players see where their boost came from.

What resets and what stays

Item	On rebirth
Money	Resets (brainstorm)
Speed	Resets (brainstorm)
Level	Open: resetting fits the milestone rule above
Car inventory	Kept (proposed): rare cars take too long to re-roll
Fuel, auras, spoilers	Open
Workshop money upgrades	Open
Jett Fuel upgrades	Open
Rebirth Points and stats	Kept
Gamepass perks	Always kept

Economy (Money)

Money comes from one place, race cash-outs, and gets spent on permanent Speed boosts.

Source: race cash-outs, multiplied by the Base Money Multi stat and the 2x Money gamepass. Idle players earn Speed but no Money, which pushes everyone to race. Confirm that's intended.

What Money buys

Purchase	Effect	Notes
Fuel types	Speed multi (F). Pricier fuel, higher multi	Open: one-time unlock, or burns over time?
Car aura	Speed multi (A) plus a visual effect	One equipped at a time (proposed)
Spoilers	Speed multi (P) plus a visual part	One equipped at a time (proposed)
Workshop upgrades	Workshop multi (W)	See workshop section
Jett Fuel upgrades	Stronger or more frequent luck bursts	See gacha section

Sample fuel ladder (placeholder numbers)

Fuel	Speed multi	Cost
Regular	1x	Free
Premium	1.5x	\$1K
Racing	2.5x	\$25K
Nitro	5x	\$500K
Plasma	10x	\$10M

If fuel, auras and spoilers reset on rebirth, players re-buy them every run and Money stays important. If they're kept, Money matters less after the first few rebirths.

Car gacha

Players auto-roll for cars for free; rarer cars carry bigger Speed multipliers, and Luck makes rare cars more likely.

Requirements

- Open gacha: rolls cost nothing and every car's odds are shown in-game (assumed meaning of "open", confirm).

- Auto-roll runs passively. Placeholder: one roll every 3 seconds.
- Base luck is 1x. Stats, rebirths, gamepasses, Server Luck and Jett Fuel raise it.
- Rolled cars go to an inventory. The player equips one car at a time.
- Rarer car, higher Speed multi.

How luck works (proposed). Each car has a "1 in N" chance. A roll checks cars from rarest to most common and awards the first hit. Luck L divides N:

$$P(\text{hit this car}) = \min\left(1, \frac{L}{N}\right)$$

So at 10x luck, a 1-in-1,000 car becomes 1-in-100.

Sample rarity table (placeholder, two cars from the brainstorm)

Car	Rarity	Base chance	Speed multi
Rusty Hatchback	Common	Catch-all (whatever's left)	1.1x
Wooden Car	Uncommon	1 in 10	1.25x
Go-Kart	Rare	1 in 50	2x
Street Racer	Epic	1 in 250	6x
Rocket Car	Legendary	1 in 1,000	25x
Hover Car	Mythic	1 in 10,000	150x
Void Racer	Secret	1 in 100,000	1,000x

Every roll must return *something*, so the table needs a catch-all common. The brainstorm's 1-in-10 Wooden Car leaves 90% of rolls undefined otherwise.

Jett Fuel (luck burst)

- Every 50 auto-rolls, the next roll gets 10x luck, stacked on top of the player's normal luck.
- Upgradeable with Money. Brainstorm example: upgraded to every 1,000 rolls for 100x luck.

Design flag: that upgrade example is a downgrade. Over 1,000 rolls, the base version gives 20 boosted rolls at 10x. The upgrade gives 1 roll at 100x. For any rare car, that's half the expected hits. Fix: upgrades should raise the multiplier *or* shorten the interval, never lengthen it.

Jett Fuel level	Triggers every	Luck on that roll	Cost (placeholder)
1 (start)	50 rolls	10x	Free
2	40 rolls	15x	\$50K
3	30 rolls	25x	\$5M
4	20 rolls	50x	\$500M

Rules for paid luck. Roblox treats paid luck boosts as paid random items ([Creator Hub](#)). That covers the luck gamepasses and Server Luck:

- Show each car's odds as a percentage, summing to 100%.
- Explain a luck purchase's effect in numbers before purchase.
- Update the displayed odds live while boosts are active.
- Check `ArePaidRandomItemsRestricted` per player and block or hide paid luck for restricted players.

Monetization

Each Robux item is either a **gamepass** (bought once, owned forever) or a **developer product** (can be bought again and again) ([Creator Hub](#)). Items that stack with every purchase can't be a single gamepass.

Item	Effect	Roblox type	Notes
2x Auto-Roll Speed	Auto-roll runs twice as fast	Gamepass	More rolls: treat as paid luck (see gacha rules)
Extra Roll I	+1 simultaneous auto-roll (2 total)	Gamepass	Same as above
Extra Roll II	+1 more (3 total)	Gamepass	Sold only after Extra Roll I
Speed Doubler	2x Speed; each buy doubles it again (2x, 4x, 8x). Price doubles each time	Chain of gamepasses (Speed I, II, III...)	Needs a cap

Item	Effect	Roblox type	Notes
2x Money	Doubles race payouts	Gamepass	
+1 Luck	+1 permanent base luck	Gamepass	Paid luck rules apply
+2 Luck	+2 permanent base luck	Gamepass	Paid luck rules apply
Server Luck	2x luck for the whole server, 15 min. Each extra buy doubles the multi, doubles the price, adds 15 min	Developer product, one per tier	Paid luck rules apply. Buyer can pause
Workshop tiers: Gold, Diamond, Void	Each tier raises the workshop multi (W)	Chain of gamepasses	Each requires the tier before

Implementation notes

- **Rising prices need one item per tier.** Each gamepass or product has one fixed price. "Price doubles each time" means separate items (Server Luck 2x, 4x, 8x) and prompting the next one.
- **Speed Doubler as a gamepass chain (recommended).** Roblox tracks pass ownership for you. A developer product would mean saving the purchase count yourself.
- **Cap the Speed Doubler.** Uncapped doubling lets one big spender outgrow every balance number. Placeholder cap: 5 buys (32x).
- **Luck passes add to base luck** before multipliers (proposed). +1 doubles a new player's luck but barely matters late game.
- **Server Luck edge cases.** Proposed: only the buyer can pause; if the buyer leaves, the timer resumes and runs out normally.

Technical and data requirements

The server decides every number (Speed, Money, rolls, race results) and the client only displays it. That's the main defense against exploiters.

- **Server-authoritative.** Compute Speed gain, payouts, rolls and race outcomes on the server. Never trust a value the client sends.

- **Workshop zone check on the server.** Check the character's position on a timer instead of trusting client-side touch events.
- **Purchases.** Grant developer products inside `ProcessReceipt` on the server. If the grant fails, return `NotProcessedYet` so Roblox retries next join. Roblox doesn't store per-player product history, so save it yourself ([Creator Hub](#)).
- **Big numbers.** Luau numbers are 64-bit floats: exact up to about 9 quadrillion, max about $1.8e308$. Stacked multipliers will get there. Build number abbreviations (1.2K, 3.4M, 5.6Qa) from day one.
- **Saving.** Use DataStores with session locking so data can't duplicate across servers; libraries like ProfileStore handle this. Save on leave, on an interval and on server shutdown (`BindToClose`).
- **Paid luck compliance.** Odds display and restricted-player checks, as listed under Car gacha.

Per-player saved data

- Speed, Money, level, XP
- Rebirth count, Rebirth Points, stat allocation
- Car inventory and equipped car
- Owned and equipped fuel, aura, spoiler
- Workshop upgrade level, Jett Fuel level, roll counter
- Developer product purchase records (receipt IDs)
- Last logout time, for offline earnings

Open questions

These decisions change what gets built. Tick them off as you decide.

Racing

- ☐ How is a race decided: speed check, win chance, hidden NPC speed or skill check?
- ☐ Is the payout a pot that grows with each win, or just the last track's reward?
- ☐ How do environments unlock: by level, or by clearing the previous environment?
- ☐ Do races get every multiplier except the workshop, or a flat base rate?

Progression

- ☐ Does level reset on rebirth?
- ☐ What survives rebirth: cars, fuel, auras, spoilers, workshop upgrades, Jett Fuel?

- ☐ Can players reset (respec) their Rebirth Points?
- ☐ Do levels unlock anything besides rebirths?
- ☐ Offline earnings: yes or no, and what percent and cap?

Economy and gacha

- ☐ Should idle players earn any Money, or only racers?
- ☐ Is fuel a one-time unlock or used up over time?
- ☐ Are workshop money upgrades and the Gold/Diamond/Void passes one track or two?
- ☐ Does "open gacha" mean free rolls with public odds?
- ☐ Inventory: size cap, duplicates, auto-delete below a chosen rarity?
- ☐ Jett Fuel upgrades: raise the multiplier, shorten the interval, or both?

Monetization

- ☐ Speed Doubler cap: how many purchases?
- ☐ Server Luck: can several players stack it at once?

Suggested build order

Build the core loop first and add layers on top, so there's something playable after every phase.

Phase	Build	Playable result
1	Starter car, workshop zone, Speed gain, saving	Idle and watch Speed climb
2	One environment, 5-track ladder, cash out or continue, Money	Core loop works
3	Fuel, auras, spoilers, workshop upgrades	Money has something to buy
4	Levels, rebirth, stat points	Long-term progression
5	Car gacha, auto-roll, inventory, Jett Fuel	Collection layer
6	Gamepasses, Server Luck, odds display, restricted-player checks	Ready to monetize

Sources

- [Paid random items policy guidelines](#), Roblox Creator Hub
- [Developer Products](#), Roblox Creator Hub
- [Remove the 20 minutes idle kick](#), Roblox DevForum